

Roll no :- S002	Name :- Kaif Ansari
Batch :- 1	Class :- SY.BSC.IT
Date of assignment :- 29/09/2026	Date of submission:-29/09/2026

PRACTICAL 30

Aim:- Cluster a dataset into two groups using k-means clustering in Breeze.

Input:-

```

build.sbt  d1.scala x
1  import breeze.linalg._
2  object d1 extends App {
3    // Dataset
4    val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)
5
6    // Initial cluster centers
7    var center1 = 2.0
8    var center2 = 10.0
9
10   // Repeat clustering
11   for (i <- 1 to 5) {
12     val cluster1 = data.toArray.filter(x =>
13       math.abs(x - center1) < math.abs(x - center2)
14     )
15
16     val cluster2 = data.toArray.filter(x =>
17       math.abs(x - center2) <= math.abs(x - center1)
18     )
19     // Calculate new centers
20     center1 = cluster1.sum / cluster1.length
21     center2 = cluster2.sum / cluster2.length
22   }
23   // Display clusters
24   println("Cluster 1:")
25   println(data.toArray.filter(x =>
26     math.abs(x - center1) < math.abs(x - center2)
27   ).mkString(", "))
28

```

```

28
29     println("Cluster 2:")
30     println(data.toArray.filter(x =>
31         math.abs(x - center2) <= math.abs(x - center1)
32     ).mkString(", "))
33
34     println("Center 1 = " + center1)
35     println("Center 2 = " + center2)
36 }
37

```

```

build.sbt x  d1.scala
1  ThisBuild / scalaVersion := "2.12.21"
2
3  libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
4
5  lazy val root = (project in file("."))
6      .settings(
7          name := "Practical 30"
8      )
9

```

Output:-

```

Run  d1 x
C:\Users\mv\luc\.jdk\openjdk-26.0.1\bin\java.exe ...
Cluster 1:
1.0, 2.0, 3.0
Cluster 2:
10.0, 11.0, 12.0
Center 1 = 2.0
Center 2 = 11.0

Process finished with exit code 0

```